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Research Paper

# Influence of standardized patient combined with narrative nursing teaching on experimental teaching of surgical nursing: A quasi-experimental study

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ABSTRACT

*Objective:* This study aimed to explore the effect of standardized patient (SP)-narrative nursing in the experimental teaching of surgical nursing.
*Methods:* A quasi-experimental study design was adopted. A total of 200 undergraduate nursing students were recruited from the Nursing College of Guilin Medical University in China from March 2023 to December 2024. The intervention group recruited students from the Class of 2022 ($n = 100$), and the control group recruited students from the Class of 2021 ($n = 100$). The intervention group adopted a teaching model combining standardized patients with narrative nursing based on traditional scenario-based simulation teaching, which was applied to the nursing of perioperative patients (4 class hours) and scenario-based case drills (4 class hours) in the experimental teaching of surgical nursing. The control group used traditional scenario-based simulation teaching. The Nurse Humanistic Care Quality Evaluation Scale, Clinical Thinking Ability Evaluation Index System Scale for Medical Students, and Nurse-Patient Communication Ability Evaluation Scale for Nursing Students were used to investigate and compare the teaching effects between the two groups of students.
*Results:* The total scores of the intervention group on humanistic care (91.39 ± 3.97), clinical thinking (79.64 ± 6.33), and nurse-patient communication (157.22 ± 7.95) abilities were significantly higher than those of the control group (82.29 ± 3.62, 65.11 ± 7.24, and 147.05 ± 7.84, respectively), with statistically significant differences ($P < 0.01$).
*Conclusion:* This study confirms that integrating the dual teaching model of standardized patients and narrative nursing in experimental teaching of surgical nursing has significantly optimized the theoretical and practical structure of teaching strategies. This innovative teaching method provides a promotable paradigm for nursing humanities education and is of positive significance for improving the effectiveness of cultivating the core literacy of nursing talents.
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## What is known?

- Standardized patients (SPs) are mainly used in medical education to help medical students improve their clinical skills and professional literacy.
- Narrative nursing refers to a nursing practice method through which nurses can better understand patients' needs and provide more personalized and compassionate care.
- Surgical nursing is a compulsory core course for nursing students.

## What is new?

- Deep integration of SPs realistic scenario simulation and the humanistic care concept of narrative nursing constructs a "technology-humanities" dual-track experimental teaching model for surgical nursing.
- The application of SPs-narrative nursing in the experimental teaching of surgical nursing can effectively improve student'

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humanistic care, clinical thinking, and nurse-patient communication abilities.
- Applying SPs-narrative nursing in the experimental teaching of surgical nursing helps create a positive classroom atmosphere.

## 1. Introduction

Surgical nursing is a core professional course focusing on practice and humanism [1,2]. In China, experimental teaching of surgical nursing is transitioning from traditional models to modern, standardized approaches. Traditional experimental teaching in surgical nursing mainly relies on scenario-based cases and medical prosthetic models and lacks innovation and simulation authenticity [3–6]. Given students' limited access to actual clinical environments and platforms for face-to-face communication with patients, they often suffer from weak humanistic care abilities, insufficient integration of theory and practice, restricted clinical thinking, and inadequate nurse-patient communication skills [7–10]. However, the application of virtual reality (VR), high-fidelity simulators, and standardized patients (SPs) in surgical nursing experimental teaching has gradually expanded. In developed countries, surgical-nursing experimental teaching demonstrates crucial advantages in frontier and systematic approaches, with high penetration of high-fidelity simulators and immersive surgical scenarios constructed via AR/VR technologies. Some U.S. institutions have introduced artificial intelligence feedback systems to analyze students' operational data in real-time and optimize guidance [11]. Finland has applied simulation games—which incorporate actual patient scenarios to facilitate effective knowledge acquisition—in surgical-nursing experimental teaching for nursing students across three universities of applied sciences [12].

SPs include individuals—healthy or with medical conditions—who undergo standardized and systematic training for the consistent and realistic portrayal of authentic clinical scenarios [13,14]. Serving three roles, they can accurately exhibit patients' clinical symptoms, signs, and medical histories: Play the role of a patient, evaluator, and teaching instructor. As a teaching method within the framework of simulation-based medical education, SPs are widely used in medical education and play an irreplaceable role in cultivating students' comprehensive clinical competencies [15,16]. Introducing SPs into teaching provides a safe, controllable, and reproducible training scenario by simulating the "patient" role in real clinical contexts. It offers a critical solution to the challenges mentioned above. SP-based teaching considerably enhances students' clinical communication skills, critical thinking, and operational standardization while reducing nurse–patient conflicts caused by operational errors, with its value widely validated in nursing education [17]. Globally, European and American systems are more mature: the U.S. incorporates SPs into medical licensure exams, and Canada has established extensive case databases. In Asia, Japan uses SPs to simulate doctor-patient conflicts, and Singapore enhances training immersion by integrating VR technology [18]. In China, SP teaching has shown rapid development: the Ministry of Education has included it in clinical medicine accreditation standards, and universities, such as Peking University and Fudan University, have established SP centers. With nursing elevated to a first-level discipline and continuous reforms in nursing education, more opportunities have emerged to introduce and apply SPs. Li et al. [19] achieved excellent teaching outcomes through scenario simulation combined with SPs in traditional Chinese medicine surgical training; meanwhile, Zhang et al. [20] observed that blended teaching with SPs further enhanced students' comprehensive clinical abilities in rehabilitation therapy education. However, compared with medical education, SPs' systematic application and research in nursing remain delayed. Current nursing education predominantly focuses on skill training and undervalues SPs' roles in humanistic care, cross-team collaboration, and complex case decision-making; it also lacks the exploration of localized practice models and concentrates mainly on courses, such as *Health Assessment*, *Medical Nursing*, and *Fundamentals of Nursing*.

Narrative nursing is an emerging model that integrates narrative medicine concepts into nursing practice. This model emphasizes that each patient has unique life stories and experiences profoundly influencing their health and illness journeys. By understanding and leveraging these stories, nurses can better address patients' needs, provide more personalized and compassionate care, cultivate students' empathy and clinical decision-making abilities, and shift their services from a "technology-centric" to a "patient-centered" paradigm [21,22]. Globally, narrative nursing teaching has developed diverse practice pathways. Western countries prioritize systematic curriculum–clinical integration: the U.S. includes "disease narrative journals" as compulsory courses, whereas the U.K. utilizes narrative reflection to enhance doctor-patient trust. Asian regions have explored localized models: Japan combines "monogatari therapy" (story therapy) to promote patients' psychological recovery, China integrate Confucian culture and family narratives into the management of chronic diseases [23]. In China has recently experienced rapid development, with certain universities offering narrative nursing as an elective course. Research has focused on its impact on students' empathy and tried to localize theories by integrating Chinese "family culture" [24]. Narrative medicine was initially proposed by Rita Charon of Columbia University in 2001 and introduced to China in 2013. Chinese nursing scholars have actively explored narrative medicine practices, extended them to narrative nursing, and continuously uncovered their humanistic appeal [25,26]. Nursing scholar Anli Jiang was among the first to introduce narrative medicine theories and gradually explore their localized application. Early research on narrative nursing concentrated mainly on clinical practice and achieved desirable outcomes, but its application in nursing education remains understudied. Huang et al. [27] applied narrative nursing in communication with ICU patients' families. They effectively promoted the establishment of nurse-patient communication models and improved family members' evaluations of communication quality. Sun et al. [28] used narrative nursing for lung cancer surgery patients; it helped them achieve posttraumatic growth and enhanced their psychosocial adaptation and quality of life.

According to the situation cognition theory, learning should occur in authentic social contexts. Traditional surgical nursing experimental teaching focuses on operational steps (e.g., wound care and drainage tube management). Still, it neglects patients' psychosocial backgrounds as Protagonist of a narrative story, which leads to mechanistic clinical decision-making by students. Charon's narrative medicine theory emphasizes that medical practice requires the ability to "absorb, interpret, and respond to patients' stories." If SPs only simulate physical symptoms (e.g., pain moaning) and ignore disease narratives (e.g., cancer-fighting experiences and family stress), students cannot train their deep communication skills. This study's innovative attempt to integrate SPs with narrative nursing in surgical nursing experimental teaching leverages their complementary strengths: SPs' realistic portrayals create authentic clinical scenario platforms where students can improve their humanistic literacy, clinical thinking, and nurse-patient communication skills in simulated real-world medical environments. This direction promotes a shift in surgical nursing experimental teaching from a "skill-centric" to a "skill-narrative integration" paradigm and provides a scalable model for

nursing education.

## 2. Methods

### 2.1. Research design and participants

A quasi-experimental research design was adopted in this study. From March 2023 to December 2024, Class of 2021 and Class of 2022, students from the undergraduate nursing program at a nursing school of a medical college in China, were selected as research participants. Based on the rationality of the historical control design (the school implements a standardized teaching management system, with curriculum settings for each grade uniformly formulated by the Academic Affairs Office, a fixed teaching faculty, and no major adjustments made to the syllabus in the past three years, and consistent classroom and experimental training facilities), the Class of 2022 students served as the intervention group and the Class of 2021 students as the control group.

The sample size was calculated using G*Power 3.1.9.7 (Heinrich-Heine-Universität Düsseldorf, Düsseldorf, Germany), with default options and the following parameters: test = *t*-tests (means: two groups), effect size = 0.5, $\alpha = 0.05$, and power = 0.8 [29]. Calculations revealed that the total sample (power = 0.8) was N1 = N2 = 51. The sample size was expanded due to the large number of nursing students in this school: Class of 2022 ($n = 100$) (intervention group) and Class of 2021 ($n = 100$) (control group). The inclusion criteria were as follows: 1) full-time enrolled Class of 2021 and Class of 2022; 2) control group students who had not participated in similar teaching reform research (to ensure the purity of historical controls); 3) students not yet in the clinical internship phase. The exclusion criteria comprised the following: 1) previous clinical probation/internship experience; 2) abnormal learning status: absence rate $\geq$15 % in the recent semester; 3) students with cognitive impairments or special educational needs. The adequate sample size of this study met the requirements. No significant differences in general information and main basic medical course scores were observed between the two classes ($P > 0.05$).

### 2.2. Intervention

Surgical nursing is one of the most essential core courses in the nursing program; it aims to cultivate students' comprehensive professional capabilities. In this study, an optimized teaching method was designed: introducing scenario-based case-SP-narrative nursing teaching into experimental teaching of surgical nursing. The experimental content mainly covered perioperative patient care (4 class hours) and scenario-based case drills (4 class hours). For the intervention group, these 8 class hours used SP-narrative nursing teaching, and the control group utilized traditional teaching methods. Both groups were educated by the same instructors, who consumed the same teaching duration and used the same textbook, *Surgical Nursing* (ISBN 978-7-117-32472-4).

#### 2.2.1. Intervention group

*2.2.1.1. Establishment of standardized patient teaching team.* Multiple teachers in this teaching team offer relevant experience and qualifications in training SPs. SPs were recruited from the school faculty, students, affiliated hospital staff, and external personnel. The SPs training process consisted of four stages: basic training, competency training, integrated training, and simulation practice. Trainees learned basic medical knowledge, in-depth case analysis, symptom and emotion performance, feedback and evaluation training, and cosmetology scenario-based practice with makeup and ultimately developed into standardized and reproducible SPs (for detailed training procedures, see Appendix A). An SP teaching team comprising five students, seven full-time teachers, three clinical teachers, and two community members was established.

*2.2.1.2. Development of narrative nursing case bank.* The construction of the narrative nursing case bank involved "story-empowered care" as its core and followed three principles: systematicness, ethics, and practicality. The teaching team divided the 44 chapters of the Surgical Nursing curriculum into seven core systems (neurosurgery, thyroid and breast surgery, cardiothoracic surgery, gastrointestinal surgery, hepatobiliary surgery, orthopedic surgery, and urology). Corresponding narrative nursing cases were developed for each system, and they were co-authored by full-time teachers and clinical teachers from affiliated hospitals based on real patients' medical conditions to ensure case authenticity and effectiveness. The narrative nursing cases were developed through the following steps. 1) Multidimensional narrative nursing material integration: typical stories were screened from clinical frontline data (e.g., ward records and nursing rounds), teaching feedback (students' reflection logs), and patient testimonials (interviews and diaries) and emphasized case diversity (age, disease type, and cultural background) and contradictoriness (e.g., treatment compliance conflicts and ethical dilemmas). 2) Ethical compliance and anonymization: patients were explicitly informed of the case usage before collection and allowed to select "complete anonymity" or "partial information obfuscation" (e.g., changing "noodle shop owner" to "self-employed"). 3) Localized narrative nursing strategy adaptation: narrative nursing language was reconstructed with regional cultures, such as using "family tree" metaphors to describe life continuation for elderly patients. 4) Teaching-practice integration design: depending on difficulty, cases were labeled hierarchically as "basic level" (e.g., communication skill demonstrations) or "advanced level" (e.g., multicultural conflict mediation) to match different teaching stages. The case bank can be further expanded by continuously adding new cases and integrating local cultural elements.

*2.2.1.3. Implementation of standardized patient-narrative nursing teaching.* Preclass preparation. Teachers distributed scenario-based cases (including preset narrative conflicts, such as Mr. Wang, a rectal cancer stoma patient, chief complaint: "The ostomy bag is a stain on my life; I'd rather die than face others") to students. Students initially divided the roles for the scenario case, with each group member portraying different roles (doctor, nurse, family member, etc.). They then collectively analyzed case materials as a team, developed solutions to case problems, and designed narrative nursing plans for the narrative conflicts in the cases (e.g., students rehearse dialogue to care for the patient: "The ostomy bag is not a stain on life but a new beginning and a fresh outlet for vitality"). During case analysis, if knowledge beyond the textbook or interdisciplinary knowledge is involved, students must independently explore and collaborate as a team to solve scenario-based problems. Meanwhile, they independently designed props and costumes for SPs based on case plot requirements.

In-class implementation. SPs "authentically recreate" clinical scenarios from cases. Students employed this "realistic" scenario platform to practice the entire nursing process: nursing assessment upon admission → nursing measures during hospitalization → health education at discharge. 1) Narrative nursing assessment (replacing traditional admission nursing assessment): SP performance: trembling, covering the face, refusing to look at the ostomy bag, whispering, "I am a monster." Student task: Identify disease

metaphors ("monster" = body stigmatization) and draw a narrative timeline: pre-illness (respected teacher) → diagnosis ("death sentence") → post-operation ("social death"). 2) Application of narrative nursing techniques: preoperative psychological nursing to intervene in the patient's anxiety; intraoperative psychological support: students presented a "life rose" prop, stating, "The stoma is the stem of the rose, allowing your life to bloom again." The SP wept after touching the prop (emotional breakthrough point). Postoperative rehabilitation training: "You once taught students to face difficulties directly; can you now lead yourself to fight the 'monster'?" The SP gradually participated in stoma care (behavioral change). 3) Narrative-oriented nursing measures: during practical ostomy bag replacement, students simultaneously guided narrative reconstruction: "Now we are cleaning the soil of the life rose—please recall your proudest class" The SP repeated teaching stories, following the students' instructions (distraction + self-empowerment). 4) Health education to reconstruct hope: students helped patients develop discharge plans through narrative methods. A narrative nursing continuation plan was designed based on the patient's identity as a teacher: "Please record a 'Rose Diary' daily: one thing accomplished because of the stoma + one act of helping others," with a "Rose Medal" gift to be pasted on the ostomy bag. Throughout this nursing process, students implemented narrative nursing for SPs, which included alleviating pre-, intra-, and postoperative anxiety and fear while practicing humanistic care skills and comprehensively analyzing nursing problems faced by patients.

Postclass reflection. Teachers and students jointly conducted in-depth retrospective reflection on the entire scenario performance process, which includes affirming students' strengths, correcting shortcomings, analyzing the achievement of teaching objectives, and requiring students to pen reflective diaries (e.g., "How did I use the 'rose metaphor' to break the patient's self-stigma?"). Reflective learning abilities were cultivated through scenario competitions (narrative nursing case competitions), with participation requirements including the submission of verbatim scripts of narrative nursing intervention dialogue and video clips of SP emotional transitions to promote learning through competition.

#### 2.2.2. Control group

The control group of students participated in traditional scenario-based case teaching with medical prosthetic models. The method involved dividing students into groups, where each group discussed different segments of the scenario case (e.g., Group 1 discussed Segment 1, Group 2 discussed Segment 2, and Group 3 discussed Segment 3). Next, the scenario case drills were completed using medical prosthetic models, and finally, teachers corrected and supplemented the students' performance in these drills.

### 2.3. Instruments

#### 2.3.1. Humanistic care ability

Liu developed the Nursing Caring Characters Assessment Tool [30] based on the theoretical model of nurses' humanistic care quality structure, which was used. The scale includes four dimensions: humanistic care philosophy (7 items), humanistic care competency (7 items), humanistic care knowledge (7 items), and humanistic care perception (8 items), totaling 29 items. The tool uses a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree), with the total score in the range of 29–145. High scores indicate good humanistic care quality. The scale demonstrated a Cronbach's α coefficient of 0.931 and a validity coefficient of 0.986.

#### 2.3.2. Clinical thinking ability

This work adopted Song's Clinical Thinking Ability Evaluation scale for medical students [31], which comprises three dimensions: critical (6 items), systematic (11 items), and evidence-based thinking abilities (7 items), with 24 items in total. A 5-point Likert scale was used, with each item scored from 1 to 5, resulting in a maximum total score of 120. For ease of interpretation, the raw scores were converted to percentage scores using the formula: Percentage score = (Raw score / 120) × 100. A higher score indicates better clinical thinking ability. The grading criteria are as follows: 80–100 points indicate excellent clinical thinking ability, 60–79 points indicate good ability, 40–59 points indicate average ability, 20–39 points indicate poor ability, and 0–19 points indicate very poor ability. The Cronbach's α coefficient of 0.91 and a test-retest reliability of 0.84 were obtained for the test.

#### 2.3.3. Nurse–patient communication ability

The Nurse–Patient Communication Ability Evaluation Scale for nursing students was developed by Xu. [32], which includes six dimensions, was used: communication planning and preparation (6 items), initiation of nurse-patient communication (7 items), information collection (11 items), information provision (6 items), gaining and understanding patient perspectives (6 items), and communication closure (6 items), totaling 42 items. The dimensions were scored using a 5-point Likert scale, with a total score range of 42–210. High scores indicate strong communication skills. The scale had an overall Cronbach's α coefficient of 0.953, with subscale coefficients ranging from 0.828 to 0.907 and a validity coefficient 0.920.

### 2.4. Data collection

Before data collection, the teachers introduced the study content, methods, and objectives to the students and conducted a pretest to analyze the rationality of the items. After the course ended for both groups, data from the Humanistic Care Ability Scale, Clinical Thinking Ability Scale, and Nurse-Patient Communication Ability Scale were collected via the online platform "Wenjuanxing" (https://www.wjx.cn/). Students completed the scales within a specified time, and data were strictly reviewed to ensure quality. In this study, the intervention group and the control group were consistent in terms of the syllabus, teaching team, and practical training conditions. To prevent students from speculating about the research purpose, data collection using the aforementioned scales was not conducted before the intervention.

### 2.5. Data analysis

Data analysis was completed using SPSS 26.0. Normally distributed quantitative data are presented as mean and standard deviation (*SD*), and intergroup comparisons were performed through independent samples *t*-tests. Categorical data were compared using chi-square tests. $P < 0.05$ indicated statistical significance.

### 2.6. Ethical considerations

This study received approval from the Ethics Committee of Guilin Medical University (Approval No. GLMC20240901), which confirmed that the data collected via the scales were utilized solely for educational research purposes and not for commercial use. All participants were informed of the study's objectives, procedures, and potential risks. In addition, they retained the right to withdraw unconditionally at any time. Data were collected

anonymously to protect privacy; no personally identifiable information was recorded.

## 3. Results

### 3.1. Participants' characteristics

A total of 200 students provided informed consent to participate in this work. The intervention group comprised 82 females and 18 males; the control group comprised 84 females and 16 males. Students in both groups were aged 19–22 years. No statistically significant differences were observed in the scores on major basic medical courses between the two groups ($P > 0.05$), which indicates that the two groups of students were comparable (Table 1).

### 3.2. Comparison of humanistic care ability between the two groups

The total score of humanistic care ability in the intervention group was (91.39 ± 3.97), and that in the control group was (82.29 ± 3.62). The difference between the two groups was statistically significant ($P < 0.01$), which implies that SP-narrative nursing teaching can effectively improve students' humanistic care ability (Table 2).

### 3.3. Comparison of clinical thinking ability between the two groups

Regarding clinical thinking ability, the intervention group had a total score of (79.64 ± 6.33), and that of the control group was (65.11 ± 7.24). The difference between the two groups showed statistical significance ($P < 0.01$), which indicates that SP-narrative nursing teaching can effectively improve students' clinical thinking ability (Table 3).

### 3.4. Comparison of nurse-patient communication ability between the two groups

For the nurse-patient communication ability, the intervention group had a total score of (157.22 ± 7.95), and that of the control group was (147.05 ± 7.84). The difference between the two groups displayed statistical significance ($P < 0.01$), which means that SP-narrative nursing teaching can effectively improve students' nurse-patient communication ability (Table 4).

**Table 1**
Comparison of baseline data among students in the two groups.

| Characteristics | Intervention group ($n = 100$) | Control group ($n = 100$) | $t/\chi^2$ | $P$ |
|---|---|---|---|---|
| Age (years) | 20.97 ± 0.85 | 20.31 ± 0.72 | 1.616 | 0.054 |
| Gender | | | 0.142 | 0.707 |
| Male | 18 (18.0) | 16 (16.0) | | |
| Female | 82 (82.0) | 84 (84.0) | | |
| Scores of major basic medical courses | 75.98 ± 5.77 | 76.38 ± 5.65 | 0.495 | 0.311 |

*Note*: Data are *n*(%) or *Mean* ± *SD*.

**Table 2**
Comparison of humanistic care abilities between the two groups of students.

| Group | Caring concept | Caring knowledge | Caring ability | Caring perception | Total score |
|---|---|---|---|---|---|
| Intervention group ($n = 100$) | 22.66 ± 2.31 | 23.88 ± 3.02 | 21.98 ± 2.58 | 22.87 ± 2.81 | 91.39 ± 3.97 |
| Control group ($n = 100$) | 19.55 ± 2.17 | 21.54 ± 2.64 | 20.76 ± 3.12 | 20.44 ± 2.56 | 82.29 ± 3.62 |
| $t$ | 9.813 | 2.792 | 3.013 | 6.393 | 16.938 |
| $P$ | <0.001 | 0.003 | 0.002 | <0.001 | <0.001 |

*Note*: Data are *Mean* ± *SD*.

**Table 3**
Comparison of clinical thinking abilities between the two groups of students.

| Group | Critical thinking | System thinking | Evidence-based thinking | Total score |
|---|---|---|---|---|
| Intervention group ($n = 100$) | 23.86 ± 5.93 | 26.23 ± 4.95 | 29.55 ± 4.07 | 79.64 ± 6.33 |
| Control group ($n = 100$) | 21.11 ± 5.61 | 21.74 ± 4.79 | 22.26 ± 3.62 | 65.11 ± 7.24 |
| $t$ | 3.369 | 6.518 | 13.384 | 15.109 |
| $P$ | 0.001 | <0.001 | <0.001 | <0.001 |

*Note*: Data are *Mean* ± *SD*.

**Table 4**
Comparison of nurse-patient communication abilities between the two groups of students.

| Group | Communication planning and preparation | Initiation of nurse-patient communication | Information collection | Information provision | Gaining and understanding patient perspectives | Communication closure | Total score |
|---|---|---|---|---|---|---|---|
| Intervention group ($n = 100$) | 22.53 ± 2.41 | 21.75 ± 2.16 | 41.93 ± 4.82 | 23.66 ± 2.86 | 22.39 ± 2.43 | 24.96 ± 2.15 | 157.22 ± 7.95 |
| Control group ($n = 100$) | 20.11 ± 2.32 | 19.23 ± 1.98 | 39.54 ± 4.56 | 22.84 ± 3.05 | 21.76 ± 2.58 | 23.57 ± 1.99 | 147.05 ± 7.84 |
| $t$ | 7.234 | 8.600 | 3.602 | 1.961 | 1.778 | 4.745 | 9.108 |
| $P$ | <0.001 | <0.001 | <0.001 | 0.030 | 0.040 | <0.001 | <0.001 |

*Note*: Data are *Mean* ± *SD*.

## 4. Discussion

By constructing an immersive teaching scenario of "SPs with narrative nursing", this study significantly improved students' competence in humanistic care, clinical thinking, and nurse-patient communication. All scores of the intervention group were significantly higher than those of the control group ($P < 0.01$), which was consistent with the results of previous studies [33–35]. These findings not only confirm the key role of situational narrative education in cultivating nursing students' humanistic literacy [36], but also reveal the deficiency of traditional nursing teaching models in the dimension of embodied cognition. In particular, they address the prevalent issue of 'theory-practice disconnection' in the current course Nurse Humanistic Literacy.

In addition, at the theoretical level, this study proposes the 'Double Helix Teaching Model for Nursing Education', which spirally integrates the empathetic gene of narrative nursing and the rational gene of clinical nursing, thereby transcending the traditional dilemma of the 'dual opposition' between humanistic qualities and skills. At the practical level, it constructs a 'dynamic' teaching system: it simulates the entire process of disease evolution through the 'time compression method' and designs 'reflective reverse teaching' to guide students in cognitive adjustment through preset errors. Through the three-dimensional interaction of 'narrative scenario - clinical decision-making - relationship construction', these mechanisms effectively promote the overall development of students' nursing competence.

In terms of nurse-patient communication competence, this study broke through the limitations of traditional training that overemphasizes language skills and standardized scripts, and constructed a three-dimensional communication competence model covering cognition, behavior, and emotion. The model includes three key components: identifying patients' health beliefs, developing micro-expression training for non-verbal communication, and designing a ladder for cross-cultural empathy development. This framework is consistent with the International Council of Nurses (ICN)'s standards for 'relational communication' and provides a quantitative basis for systematic communication teaching and assessment.

At the level of clinical thinking training, based on the situated cognition theory [37], this study achieved two important improvements by applying dynamic narrative nursing cases to the experimental teaching of surgical nursing: establishing a real-time feedback link of 'symptom evolution - nursing assessment - intervention adjustment'; and introducing the situational simulation teaching segment of 'clinical medical errors'. It effectively made up for two major limitations of traditional experimental teaching of surgical nursing, which relies on static models: the inability to reproduce the dynamics of clinical decision-making and the lack of reflective scenarios for clinical errors [38]. This not only enhanced students' clinical reasoning ability, but also promoted the development of their adaptive expertise, which is of great value in shortening the clinical adaptation period.

In terms of cultivating humanistic care, this study achieved three innovations by integrating narrative nursing cases into nursing teaching: transforming disease narratives from textual symbols into concrete situational experiences; establishing students' emotional labor regulation mechanism through the emotional feedback of SPs; and developing a three-dimensional training pathway of 'narrative reflection - situational practice - ethical decision-making' [39]. This framework aligns with the WHO's nursing education reform direction of 'patient experience-centered' [40], provides a practical solution to address the formalization dilemma in humanistic education, and plays a positive role in improving patient experience, promoting nurse-patient harmony, advancing the development of the nursing profession, and facilitating the construction of 'Healthy China' [41–43].

## 5. Limitations

The study's participants were nursing students from a medical college in China, with a small number of SPs—particularly those without a medical background—and a limited number of narrative nursing cases. Therefore, future efforts should focus on cultivating high-quality nonmedical SPs to enrich the narrative nursing case bank for surgical nursing courses and expand the sample to include students from other universities or disciplines.

## 6. Conclusion

This study verifies the effectiveness of the integrated "SPs with narrative nursing" model in the experimental teaching of surgical nursing, and demonstrates the significant advantages of situational teaching from three dimensions: humanistic care, clinical thinking, and nurse-patient communication. This model provides a key path for the transformation of nursing education paradigms and holds certain academic prospectiveness and practical innovation value. Future research can advance in three directions: first, explore the integration of VR with the "SPs with narrative nursing" model, and construct a "digital and intelligent ward" to achieve ultra-realistic simulation; second, conduct longitudinal tracking to establish a "humanistic literacy attenuation model" for nursing students; third, promote the penetration of narrative nursing from education to nursing quality management, and explore its potential as a clinical early warning indicator. These explorations are expected to further expand the value boundary of nursing humanistic education.

## CRediT authorship contribution statement

**Wenshuai Tang:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Visualization, Project administration, Funding acquisition. **Yan Gao:** Conceptualization, Methodology, Validation, Resources, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

## Data availability statement

The datasets generated during and/or analyzed during the current study are not publicly available due to (this study was approved by the Ethics Committee of Guilin Medical University [Approval Number: GLMC20240901], and data sharing requires review by this Committee) but are available from the corresponding author upon reasonable request.

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## Declaration of competing interest

The authors declare there is no conflict of interest.

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## Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.06.011.

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